

Global Energy Storage

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Battery Weekly 8 June

America

- [SK On Unveils Next-Generation 'GRIDON' ESS in US.- thelec.net](#) SK On unveiled its next-generation ESS platform “GRIDON Gen2” at CLEANPOWER 2026, targeting mass production in Q3 2027. The new system supports both DC and AC configurations, with around 15% higher energy capacity per container and enhanced safety features, including EIS monitoring and liquid-based fire suppression. Leveraging its ~100 GWh U.S. production capacity, SK On aims to secure more than 20 GWh of ESS orders in 2026, with over 10 GWh already under discussion with U.S. customers, strengthening its push into the North American ESS market.
- [Rocket Battery Maker Sebang Battery to Invest 100 Billion Won in US ESS Business.- thelec.net](#) Sebang Battery plans to invest KRW 100 billion (~USD 72 million) to expand into the U.S. energy storage (ESS) market, by funding its new subsidiary Sebang Lithium Battery America. The investment—supported by the founding family through a capital increase—will establish a dedicated North American ESS business unit. The move aligns with Sebang’s strategy to pivot from traditional lead-acid batteries toward ESS and eco-friendly energy solutions, while leveraging favorable U.S. policies and strong projected growth in the ESS market.
- [Old Waymo Batteries Will Get A Second Life As Stationary Energy Storage.- insideevs.com](#) Waymo has partnered with B2U Storage Solutions to repurpose its retired robotaxi batteries into grid-scale energy storage systems. The project will deploy hundreds of megawatts of second-life battery capacity, starting in Texas and California, to store excess renewable energy and support grid demand. The initiative reflects a growing industry trend where used EV batteries are reused for stationary storage, extending their lifecycle and reducing the need for immediate recycling while supporting renewable energy integration.
- [Moment Energy opens new second-life facility in Vancouver.- elective.com](#) Moment Energy is launching a second-life battery facility in Vancouver to repurpose used EV batteries into stationary energy storage systems (BESS), with expected annual capacity of 1 GWh. Instead of recycling for raw materials, the company reuses batteries from sources such as Mercedes-Benz, extending their lifecycle for applications like data centers, industrial users, and utilities. The facility, supported by over USD 100 million in total funding, reflects growing demand for second-life battery solutions to support grid storage and renewable integration.
- [Graphite deal with Tesla: Syrah reports resolution in delivery dispute.- elective.com](#) Syrah Resources has resolved a key dispute with Tesla over a graphite supply agreement, with Tesla withdrawing its termination notice after accepting that Syrah is now producing compliant battery-grade anode material. The deal involves supplying 8,000 tonnes of graphite annually from Syrah’s Vidalia, Louisiana plant, but final contract approval remains pending as qualification tests are still ongoing. The development reduces short-term uncertainty but highlights ongoing risks in non-Chinese battery material supply chains, as Syrah aims to become a major global graphite supplier.

Asia

- [China's Tinci Inks New Deal With Major Client to Double Battery Materials Order.- yicai.com](#) Tinci Materials has signed an expanded agreement with Cornex New Energy to supply over 1.01 million tons of battery electrolytes and chemicals by 2030, nearly doubling the original contract. The deal reflects strong growth in energy storage battery demand, as Cornex rapidly expands capacity to 180 GWh annually, including a new 50 GWh ESS-focused plant. The agreement strengthens Tinci’s position as a leading global electrolyte supplier and highlights accelerating scale-up across the battery supply chain.

- [China's AI Computing Boom Tests the Limits of Green Power, Experts Say- yicaiglobal](#) China's rapid growth in AI computing demand is exceeding the pace of renewable energy deployment, creating challenges in aligning clean power supply with data center needs. Experts warn that despite strict efficiency and renewable targets, 24/7 green power matching remains difficult, with gaps between reported renewable usage and actual grid conditions. Rising AI workloads are driving sharp increases in electricity demand and power instability risks, highlighting the need for better coordination between computing workloads, grid infrastructure, and energy storage solutions such as multi-hour batteries and flexible load management.
- [EcoPro to Pursue Company-Wide AI Transformation by 2028.- thelec.net](#) EcoPro plans to implement a company-wide AI transformation (AX), aiming to become a fully AI-driven enterprise by 2028. The strategy includes deploying AI across R&D, manufacturing, and operations, reducing product development time by up to 50%, improving productivity by 30%, and enhancing defect detection accuracy to 95%. The company also plans to build autonomous factories and laboratories, using AI and robotics to optimize efficiency, lower energy consumption, and enable 24/7 automated operations.
- [ITM Semiconductor Joins Government Project to Develop Battery Thermal Runaway Delay Technology.- thelec.net](#) ITM Semiconductor is participating in a South Korea government-backed project (2026–2030) to develop technologies that delay thermal runaway propagation in pouch-type batteries. The project focuses on combining pressure pad structures and graphene-based thermal management to suppress heat spread and improve safety. ITM will be responsible for heat dissipation design, fire propagation prevention, and system validation, with applications across EVs, ESS, robotics, and defense.
- [Mintech Launches 'Mintech Ionics' for Next-Generation Battery Development.- thelec.net](#) Mintech has established a new subsidiary, Mintech Ionics, to focus on next-generation energy materials, including solid-state battery electrolytes, anodes, fuel cell components, and solar materials. Leveraging its expertise in battery diagnostics (EIS technology), the company aims to accelerate material development and improve ion transport performance. The move expands Mintech's role from diagnostics into core material supply, positioning it for growth in solid-state batteries, hydrogen, and solar energy markets.
- [IBK to launch new fund to promote renewable-energy equipment.- pulse.mk](#) IBK (Industrial Bank of Korea) plans to launch a KRW 250 billion (~USD 167 million) fund to support the use of locally produced renewable energy equipment and strengthen domestic supply chains. The fund will provide low-interest loans and equity support to projects using Korean-made technologies, helping local manufacturers compete with imports. It is part of IBK's broader strategy to expand energy infrastructure financing, including BESS, grid, and data center projects, as the bank targets KRW 8 trillion in energy investments over five years.
- [HyperStrong Secures Over 1 GWh of Energy Storage Orders in Southeast Europe.-solarbe.com](#) HyperStrong has secured over 1 GWh of energy storage orders in Southeast Europe, including more than 900 MWh of large-scale grid storage projects in Romania and an additional 75 MWh of utility-scale and commercial & industrial (C&I) systems in Croatia and Serbia, supporting applications such as grid stability, renewable energy integration, and power demand management. The deals mark a significant step in the company's European expansion, strengthening its presence in rapidly growing Balkan markets where demand for large-scale and diversified energy storage solutions is accelerating.
- [Wasion Energy Storage Secures First EU Battery Regulation Certification for Containerized ESS.-solarbe.com](#) Wasion Energy Storage has become the first company globally to receive the EU Battery Regulation (EU 2023/1542) NB certification for its GridUltra liquid-cooled containerized energy storage system, marking a key milestone for entering the European market. The certification confirms compliance with stringent EU requirements on battery safety, lifecycle management, and environmental standards, effectively serving as a "passport" for Europe. The GridUltra system (3.3–5.0 MWh) features advanced liquid-cooling and integrated safety design, highlighting Wasion's capability to meet high regulatory standards and strengthening its competitiveness in the European utility-scale ESS market.
- [Xingchen Energy and Chuneng Sign 10 GWh Battery Strategic Partnership.-solarbe.com](#) Xingchen Energy and Chuneng New Energy have signed a 10 GWh lithium-ion battery strategic cooperation agreement to strengthen collaboration across supply chain coordination, product integration, quality control, and project delivery. The partnership aims to support Xingchen's rapid expansion in large-scale energy storage projects by improving supply reliability, production capacity alignment, and delivery efficiency. Leveraging Chuneng's manufacturing and scaling capabilities, the deal enhances Xingchen's ability to execute multi-region, multi-project ESS deployments, reinforcing its industrial ecosystem as the market shifts from capacity expansion toward operational efficiency and value creation.

Europe

- [Cylib and Vianode im to Close the Battery Graphite Loop.- electrive.com](#) Cylib and Vianode have signed a collaboration agreement to develop recycled graphite for next-generation battery anodes, aiming to strengthen Europe's circular battery supply chain. Cylib will use its OLic recycling technology to recover high-purity graphite from used batteries, while Vianode will integrate and test the material in anode

production and pilot projects. The partnership aims to reduce reliance on imported raw materials and lower CO₂ emissions, supporting Europe's push for a more sustainable and locally sourced battery ecosystem.

- [Honda invests in battery materials manufacturer Nexeon.- electrive.com](#) Nexeon, a UK-based developer of silicon anode materials, has received a new investment from Honda through its Honda Xcelerator Ventures program to support further growth and commercialization. The funding will accelerate development of silicon-based anodes, which can significantly improve energy density and charging performance compared to traditional graphite. Nexeon is scaling production at its Gunsan, South Korea facility, targeting large-scale output and supplying customers such as Panasonic, highlighting growing industry interest in next-generation battery materials.
- [ProLogium targets SPAC merger to fund French battery plant.- electrive.com](#) ProLogium plans to merge with SPAC TDAC (Translational Development Acquisition Corp.) to raise at least USD 250 million and accelerate the commercialization of its solid-state batteries. The deal will enable ProLogium to list on Nasdaq (ticker: PRLG) and fund construction of its gigafactory in Dunkirk, France, which is scheduled to start production in 2028 with initial capacity of 0.8 GWh, expanding to 12 GWh by 2032. The move provides faster access to capital markets and supports scale-up of its next-generation lithium-ceramic solid-state technology for EVs and other applications such as data centers and aerospace.
- [Samsung SDI reportedly to supply battery cells to Volkswagen.- electrive.com](#) Samsung SDI is reportedly set to become a new supplier for Volkswagen's standardized "Unified Cell" battery platform, alongside PowerCo and Gotion. Samsung SDI is expected to convert production lines at its Hungary plant to manufacture the prismatic cells, with mass production starting around 2027 and capacity reaching double-digit GWh levels. The move would strengthen Volkswagen's battery supply chain in Europe and support its flexible platform strategy, which can use multiple chemistries such as LFP, NMC, and future solid-state batteries.
- [Europe's Largest Flow Battery Project \(800 MW/1.6 GWh\) Begins Construction.- solarbe.com](#) Flexbase has started building an 800 MW / 1.6 GWh flow battery project in Switzerland, the largest in Europe, located at a key cross-border grid hub. The project combines long-duration energy storage, an AI data center, and district heating, forming a zero-carbon energy system where renewable power is stored, used for computing, and waste heat is reused for heating. It is expected to start operation in 2028 and significantly support grid stability and decarbonization.

EXHIBIT 1: **Key commodities price performance**

	Price			EV						
	5-Jun	Ccy	Mkt cap	P/E	/EBITDA	%1D	%1W	%1M	%1Y	1Y px
			\$bn	26E	26E					chart
Metal prices (US\$/tonne)										
LiCO (spot)	24,580					-2.9%	-4%	-3%	193%	
LiOH (spot)	24,506					0.1%	0%	1%	186%	
LiCO (contract)	20,250					0.0%	0%	8%	108%	
LiOH (contract)	20,000					0.0%	0%	8%	135%	
Cobalt (spot)	55,858					0.0%	0%	0%	68%	
Nickel (spot)	18,689					-1.0%	-2%	-5%	20%	
Component prices										
Cathode NMC811						0.0%	0%	3%	53%	
Cathode LFP						#N/A	#N/A	#N/A	#N/A	
Precursor NMC811						#N/A	#N/A	#N/A	#N/A	
Precursor LFP						0.0%	2%	8%	32%	
Artificial Graphite						0.0%	0%	2%	4%	
Natural Graphite						0.0%	0%	0%	-6%	
Electrolyte NMC						0.0%	0%	-3%	23%	
Electrolyte LFP						0.0%	0%	-3%	43%	
Separator						0.0%	0%	5%	17%	
Spot battery cell cost (US\$/kWh)										
LFP	44					0.0%	0%	0%	4%	
NMC532	90					0.0%	0%	2%	30%	
NMC622	89					0.0%	0%	2%	27%	
NMC811	88					0.0%	0%	1%	25%	
Spot battery pack cost (US\$/kWh)										
LFP	59					0.0%	0%	-2%	-9%	
NMC532	122					0.0%	0%	2%	21%	
NMC622	121					0.0%	0%	2%	19%	
NMC811	121					0.0%	0%	1%	17%	

Source: Bloomberg, Baiinfo, Bernstein analysis and estimates (cell and pack cost)

EXHIBIT 2: Key companies price performance and valuation

	Price				EV						
	5-Jun	Ccy	Mkt cap	P/E	/EBITDA	%1D	%1W	%1M	%1Y	1Y px	
			\$bn	26E	26E					chart	
Cell makers											
LGES	408,000	KRW	61.8	755.5x	16.7x	-1.2%	-11%	-13%	40%		
Samsung SDI	559,000	KRW	29.2	144.5x	19.3x	-5.1%	-19%	-20%	221%		
CATL	408.2	CNY	285.8	19.6x	13.9x	-4.3%	-4%	-6%	62%		
CALB	28.7	HKD	7.0	17.2x	5.1x	-6.1%	-2%	-19%	71%		
EVE Energy	59.7	CNY	19.1	17.6x	11.3x	-5.3%	-6%	-18%	29%		
Gotion	30.7	CNY	8.2	25.7x	7.5x	-3.7%	-7%	-17%	23%		
Sunwoda	21.2	CNY	5.8	15.2x	5.8x	-2.9%	-7%	-23%	14%		
Farasis	12.1	CNY	2.2	n.a.	n.a.	-2.7%	-7%	-13%	-8%		
Lithium Miners/Refiners											
Albemarle	165.7	USD	19.5	13.3x	6.8x	-1.6%	-6%	-15%	182%		
SQM	77.7	USD	21.3	11.1x	6.3x	-1.1%	-10%	-16%	141%		
Pilbara Minerals	5.9	AUD	13.6	34.5x	17.4x	-3.3%	-8%	-2%	340%		
Ganfeng Lithium	57.6	HKD	19.5	13.9x	12.1x	-4.9%	-10%	-33%	188%		
Tianqi Lithium	59.5	CNY	14.3	16.1x	5.4x	-2.5%	-5%	-26%	96%		
Cathode											
LG Chem	331,500	KRW	15.1	145.8x	3.0x	-2.8%	-10%	-22%	65%		
EcoPro BM	184,200	KRW	11.7	403.4x	67.6x	-4.7%	-15%	-14%	91%		
L&F	137,800	KRW	3.6	639.4x	20.3x	-5.9%	-18%	-36%	143%		
Umicore	24.3	EUR	7.0	14.9x	6.2x	-4.2%	-4%	20%	145%		
Ningbo Ronbay	32.4	CNY	3.4	52.6x	14.9x	3.3%	4%	-7%	75%		
Beijing Easpring	52.6	CNY	4.2	26.6x	17.1x	-2.2%	-5%	-17%	36%		
GEM	7.4	CNY	5.6	17.5x	5.6x	-3.8%	-5%	-19%	21%		
Huayou Cobalt	51.3	CNY	14.4	10.9x	5.2x	-3.6%	-6%	-23%	50%		
Dynanonic	61.8	CNY	2.6	28.0x	10.1x	-3.5%	-6%	-14%	118%		
Anode											
Posco	205,500	KRW	11.8	323.8x	50.6x	-5.5%	-16%	-23%	78%		
Putailai	28.4	CNY	9.0	18.3x	11.5x	-5.2%	-7%	-22%	77%		
Shanshan	13.6	CNY	4.5	n.a.	n.a.	-2.2%	-7%	-18%	64%		
Separator											
Asahi Kasei	1,784	JPY	15.2	16.6x	6.0x	-0.1%	0%	17%	81%		
SK IE	18,040	KRW	1.0	n.a.	-40.2x	-2.1%	-7%	-33%	-19%		
Toray Industries	1,112	JPY	10.5	20.7x	6.4x	0.5%	-7%	0%	14%		
W-Scope	308	JPY	0.1	n.a.	n.a.	-1.9%	7%	34%	6%		
Yunnan Energy	64.4	CNY	9.3	36.6x	13.2x	-3.0%	-9%	-22%	131%		
Shenzhen Senior	15.6	CNY	3.1	34.4x	10.6x	-0.8%	-2%	-5%	55%		
Sinoma Science	63.8	CNY	15.8	39.5x	15.8x	0.3%	-10%	16%	272%		
Electrolyte											
Enchem	33,750	KRW	0.5	n.a.	n.a.	-2.6%	-5%	-25%	-51%		
Tinci Minerals	49.4	CNY	14.9	14.9x	10.8x	-3.7%	-5%	-18%	185%		
Capchem	78.3	CNY	8.7	28.8x	18.2x	10.2%	6%	13%	153%		
Jiangsu Guotai	8.3	CNY	2.0	10.6x	n.a.	-4.4%	-7%	-7%	12%		
Shanshan	13.6	CNY	4.5	n.a.	n.a.	-2.2%	-7%	-18%	64%		
Equipment Maker											
Hirano	2,244	JPY	0.2	21.2x	n.a.	-4.8%	12%	25%	54%		
CKD	6,200	JPY	2.6	33.3x	17.1x	-1.6%	-5%	3%	169%		
Wuxi Lead	46.9	CNY	11.4	31.2x	24.7x	-1.7%	-8%	-15%	134%		
Shenzhen Yinghe	24.7	CNY	2.4	26.2x	12.8x	-1.6%	-9%	-15%	41%		
Zhejiang HangKe	33.9	CNY	3.0	35.6x	33.5x	-4.0%	-15%	6%	97%		
Guangdong Lyric	50.4	CNY	1.3	55.4x	20.3x	-2.6%	-12%	-7%	72%		
Shenzhen United	26.1	CNY	1.3	23.4x	16.0x	0.7%	-6%	-9%	51%		

Source: Bloomberg (all estimates), Bernstein analysis
GLOBAL ENERGY STORAGE

Company Models

[CATL](#) | [LG Energy Solution](#) | [Samsung SDI](#)

[LG Chem](#) | [EcoPro BM](#) | [Posco Future M](#)

[Tianqi Lithium](#)

[Sungrow](#)

Industry Models

[Model: Monthly EV Tracker](#)

[Model: Battery Gigafactory Database](#)

[Industry model: Global EV TAM](#)

[Industry model: Global ESS TAM](#)

[Industry model: Battery TAM](#)

[Industry model: Battery supply and demand](#)

Blackbook

[Battery Utopia](#)

[Battery Tech Roadmap 2023: Faster, Higher, Stronger](#)

[Turn Up the Voltage... Investing Across the Battery Value Chain](#)

[Global Energy Storage: Resistance is Futile](#)

Initiation Report

[Global Energy Storage: Initiating Coverage. Lithium Positive \(Tianqi\) and Negative Cathodes \(EcoPro BM, Posco Future M\)](#)

[Sungrow: Solar Eclipse - Initiating at Market-Perform \(PT RMB65\)](#)

Recent Research

[Global Energy Storage: Battery takeaways from Beijing Auto Show](#)

[EV TRACKER - 2025: Solid year of 20% growth, but expectations are for slower 2026](#)

[CATL: A winner in the age of energy disruption](#)

[Future of Tech: Can China achieve AI supremacy?](#)

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[The Long View: Batteries to only get better](#)

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[Tianqi Lithium: 1Q26 Best Idea - Energy storage powers lithium upside. Outperform](#)

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[Global Energy Storage: 2026 Outlook. Power storage set to continue to drive the battery rally](#)

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[Global Energy Storage: Should investors chase the rally in battery stocks?](#)

[Global Energy Storage: Why is ESS demand booming?](#)

[Global Energy Storage: Increasing CATL price target to CNY420 on blockbuster year for ESS](#)

[China EV & Battery Tour Takeaways, 2025 Edition: Steady Evolution](#)

[Global Energy Storage: Can US ESS demand offset EV weakness for Korean battery makers](#)

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[Global Energy Storage: Has a Deepseek moment arrived for solid state batteries?](#)

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[CATL: 2025 Earnings Preview. So far so good.](#)

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[Future of Tech: Lift off for the low-altitude economy](#)

[Global Energy Storage: How big is the datacenter opportunity for batteries?](#)

[Global Energy Storage: More downside risks ahead for Korean cathode makers](#)

[CATL: Does CATL deserve a premium or discount for HK IPO?](#)

[CATL: Initial perspectives on HK IPO Plans](#)

[Global Energy Storage: Energy storage demand continues to surge on renewables and data centers](#)

[Global Energy Storage: Gigafactory expansion still strong but investment slowing](#)

[Global Energy Storage: Bigger and better batteries enables the rise of the 700km electric vehicle](#)

[Global Energy Storage: 2025 Lithium Outlook. Rising from rock bottom](#)

[Global Energy Storage: Outlook 2025. New markets to drive battery growth](#)

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	4 Jun 2026	Price	TTM	Reported EPS				Reported P/E (x)		
			Closing	Target	Rel.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
300750.CH (CATL)	O	CNY	408.20	620.00	22.0%	CNY	16.14	21.56	26.02	25.3	18.9	15.7
3750.HK (CATL)	M	HKD	723.50	600.00	94.1%	CNY	16.14	21.56	26.02	38.8	29.1	24.1
051910.KS (LG Chem)	M	KRW	341,000	298,000	27.1%	KRW (13,258.70)	2,042.47	31,118		(25.7)	167.0	11.0
373220.KS (LGES)	M	KRW	413,000	347,000	(0.2)%	KRW (5,308.10)	1,811.04	9,452.97		(77.8)	228.0	43.7
006400.KS (SDI)	M	KRW	589,000	520,000	195.8%	KRW (9,933.80)	2,099.51	18,376		(59.3)	280.5	32.1
003670.KS (Posco Future M)	U	KRW	217,500	190,000	46.5%	KRW (2,740.96)	376.72	898.20		(79.4)	577.3	242.2
247540.KS (EcoPro BM)	U	KRW	193,300	140,000	58.8%	KRW	403.00	854.00	1,963.00	479.7	226.3	98.5
300274.CH (Sungrow)	O	RMB	164.35	185.00	125.8%	RMB	6.55	8.22	9.33	25.1	20.0	17.6
002466.CH (Tianqi Lithium)	O	CNY	59.46	73.00	54.4%	CNY	0.28	3.29	6.18	210.7	18.1	9.6
9696.HK (Tianqi Lithium)	O	HKD	45.76	61.00	31.6%	CNY	0.28	3.29	6.18	140.5	12.1	6.4
ASIA			1,975.13									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

003670.KS base year is 2024;

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

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- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
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